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Экономика суперзвёзд в условиях жёсткого потолка зарплат: исследование Национальной Хоккейной Лиги

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Economics of Superstars under Hard Salary Cap: a Study of the National Hockey League

Аннотация

Суперзвёзды могут значительно усиливать выступление своей команды, но в то же время они увеличивают неравенство зарплат среди игроков, которое может уменьшить положительный эффект суперзвёзд и даже сделать невыгодным их найм. В моей работе я моделирую этот трейд-офф, фокусируясь на командах Национальной Хоккейной Лиги (НХЛ), действующих в условиях потолка зарплат. Используя данные по зарплатам игроков НХЛ, я оцениваю коэффициенты модели, и она предсказывает, что для таких значений параметров командам следует нанимать суперзвёзд. Далее я исследую возможную гетерогенность эффектов и нахожу, что лучшие игроки менее восприимчивы к отрицательным эффектам от неравенства и выигрывают сильнее от наличия суперзвёзд в команде по сравнению с остальными игроками. Более того, суперзвёзды могут незначительно ухудшать выступление игроков, с которыми они конкурируют за время на льду, но эффект на игроков, выступающих на других позициях, строго положительный. Основываясь на этих результатах, командам НХЛ следует нанимать суперзвёзд, но при этом не создавать конкуренцию между ними.

Abstract

The superstars who can boost the performance of their team, at the same time, can increase the inequality that may mitigate the positive effect of superstars and even make hiring them unfavorable. In my thesis, I model this trade-off by focusing on the National Hockey League (NHL) teams that operate under a hard salary cap. Then I estimate the parameters of the model using the data on the salaries of the NHL players, and the model suggests that teams should hire superstars under these conditions. Further, I study the potential heterogeneity of the effects and find out that top players are less vulnerable to adverse inequality effects and benefit from the presence of superstars more than other players. Moreover, I get that superstars may slightly decrease the performance of their peers who compete with them for time on ice, but the effect on the players with other roles is strictly positive. Based on these results, the NHL teams should hire the superstars but not create competition among them.

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1 Introduction

”I don’t question their dedication, but I do question just our ability as a group to get it done in those difficult times. And that’s why we have to make some changes...” — Brendan Shanahan, the President of the Hockey Team Toronto Maple Leafs, May 2024¹

The presence of superstars, small numbers of people who earn vast amounts of money and dominate the activities in which they engage, is a common thing for many kinds of sports, including hockey. However, whether it is optimal for the teams to hire these superstars when they have a limited budget is an unanswered question. Hence, many teams tried different strategies in search of success. In recent years, the Toronto Maple Leafs, one of the top National Hockey League teams, have been following the strategy of hiring superstar attacking players, expecting that they will lead the team to the championship and victory in the Stanley Cup. Nevertheless, year after year, they failed to achieve their goal, and the latest defeat in the first round of the playoff series of the Stanley Cup forced the club’s management to question their strategy of hiring superstars, which is what the quote at the beginning of this paper is dedicated to. Furthermore, this loss reheated the ongoing discussion about the need to have superstars on the team. In my thesis, I address this question by developing a theoretical framework, demonstrating the potential pros and cons of having superstars, and estimating these positive and negative effects empirically.

To the best of my knowledge, the first paper that discusses the phenomenon of superstars and the reasons why they exist in the markets is the seminal paper by [Rosen \(1981\)](#). In this study, he defined superstars as a small number of people dominating the activities in which they engage and earning enormous amounts of money, and he showed why their presence in sports is not a surprise. After that, there emerged a special field exploring the potential positive effects of superstars on their teams and the leagues they participate in. For instance, [Hausman and Leonard \(1997\)](#) discover that superstar players in the National Basketball Association have a positive effect not only on the revenues of the team they are playing for but also on the revenues of their opponents. Additionally, [Deutscher and Schneemann \(2017\)](#) claim that there is a positive effect of superstars on the attendance of hockey games in the German and Czech leagues. Speaking of the impact on teams’ productivity and performance, the results obtained by [Cyrenne \(2018\)](#) suggest that the higher the maximum wage paid by a team, the better the performance of an NHL team. This result is robust to different specifications of the models, which can mean that superstars boost their teams’ performances.

Nonetheless, the teams cannot simply make a squad consisting only of superstars because of the limited budget. Therefore, their squads usually consist of players of different levels of talent and wages, which can affect the productivity of the teams. However, the

¹The quote is taken from Shanahan’s interview with ESPN ([Shilton, 2024](#))

direction of this effect is not obvious. On the one hand, large differences in wages in the team may encourage the players to make more effort to obtain the higher salaries, which was shown by the example of tournament contestants in [Lazear and Rosen's](#) paper (1981). On the other hand, high wage dispersion may have a negative effect on the team's performance, especially in cases where cohesion is required. This idea is developed in [Levine's](#) theoretical model (1991).

Many researchers tried to test which of the previously mentioned effects dominates in professional sports, and they again got mixed results depending on the kind of sport. For example, in Major League Baseball, many researchers found a negative effect ([Richards and Guell, 1998](#); [Depken II, 2000](#); [Jewell and Molina, 2004](#); [Debrock et al., 2004](#)). However, the common opinion is that there is no effect of inequality on the performance of the teams in the National Basketball Association ([Berri and Jewell, 2004](#); [Katayama and Nuch, 2011](#); [Simmons and Berri, 2011](#)). Speaking of the National Hockey League, the most recent studies claim that the inequality effect is negative, but sometimes it appears to be statistically insignificant ([Stefanec, 2012](#); [Depken and Lureman, 2018](#); [Cyrenne, 2018](#)).

Thus, there are plenty of papers showing the presence of either the superstar effect or the inequality effect. However, none of these papers tries to properly estimate both effects simultaneously and conclude whether it is optimal to have superstars on a team or not. Moreover, the papers studying the effects of the National Hockey League mainly use data on the seasons from the 2000s, so little is known about the situation in modern hockey. Finally, most of the papers use team-level data for their analysis, which does not allow them to estimate the potential peer effects of superstars on their teammates.

In my thesis, I try to address all these issues. First, I develop a theoretical model in which I model the non-negative superstar effect and the non-positive inequality effect, which is consistent with what the existing literature suggests about the NHL. In the beginning, I solve this model for arbitrary parameters and then estimate the parameters using real-world data. For these estimations, I use player-level data that covers all seasons from 2011/12 to 2022/2023, which allows me to get a much larger and more relevant dataset relative to the previously mentioned studies. Using this data, I estimate the parameters of the model and find that under these parameters, the teams benefit from hiring superstars. However, the estimated optimal number of superstars is higher than the actual number that the teams usually hire. Nevertheless, I attribute it not to the non-optimal decisions made by teams but to the limited supply of superstar players.

Additionally, the player-level data allows me to study the peer effects of the players and the heterogeneity of the effects. I find that the top players are less vulnerable to the potential negative effects of inequality and benefit more from the presence of superstars. Furthermore, having superstars differently affects players with different roles. For example, defenders benefit from good attacking players, but many good defenders may decrease the performance of other defenders through increased competition. Similar re-

sults are observed for the attackers who benefit from superstar defenders but may suffer from competition with superstar attackers.

My paper contributes to several strands of literature. First, it relates to the literature on superstar effects in different kinds of sports. I find the positive effect of superstars in the National Hockey League, which is in line with the results in existing literature (Cyrenne, 2018). Additionally, I develop the theoretical model that allows me to show that the benefits of having superstars exceed the potential disadvantages, which makes hiring superstars optimal in the modern NHL.

Second, this paper may be attributed to the literature studying peer effects. Many papers find the positive effects of superstars on their peers in team sports (Depken and Haglund, 2011; Molodchik et al., 2021). However, in individual sports, the superstar peer effect is often negative (Brown, 2011; Deutscher et al., 2023). In my paper, I show that the players benefit from having superstars in other positions. At the same time, superstars may negatively affect their peers who have the same role, which is consistent with the results for individual sports when sportsmen compete with superstars. Overall, similar results were obtained by Gould and Winter (2009) for Major League Baseball. They find that a player's effort has a positive effect on the effort of his peers if they are complements in production and a negative effect if they are substitutes. Thus, my paper provides some evidence for the results of this study by focusing on another setting.

Finally, I contribute to the literature studying how inequality affects productivity. I find the evidence in favor of Levine's theory (1991) in the NHL setting. Similar results were obtained in the previous studies focusing on the earlier time periods (Stefanec, 2012; Depken and Lureman, 2018; Cyrenne, 2018), so I show that the effect is persistent and can be observed even in the latest seasons.

The rest of the paper proceeds as follows. Section 2 describes the NHL setting. Section 3 focuses on the main assumptions and results of my theoretical model. Section 4 presents the data. Section 5 discusses the empirical strategy. Section 6 presents the results, and Section 7 concludes.

2 Setting

In this section, I briefly describe the main rules of the National Hockey League that are essential for understanding my further analysis.

To begin with, the National Hockey League is a tournament consisting of two parts: the regular season and the Stanley Cup playoff. During the regular season, each team plays 82 games. If it wins the game, it gets 2 points. If it loses in overtime or shootout, it gets 1 point, and the team losing by the end of regulation time gets 0 points. Then these points are summed up, and the best teams, based on the number of points scored, get into the playoffs of the Stanley Cup, where they try to win the trophy.

Table 1: The history of the salary cap's changes

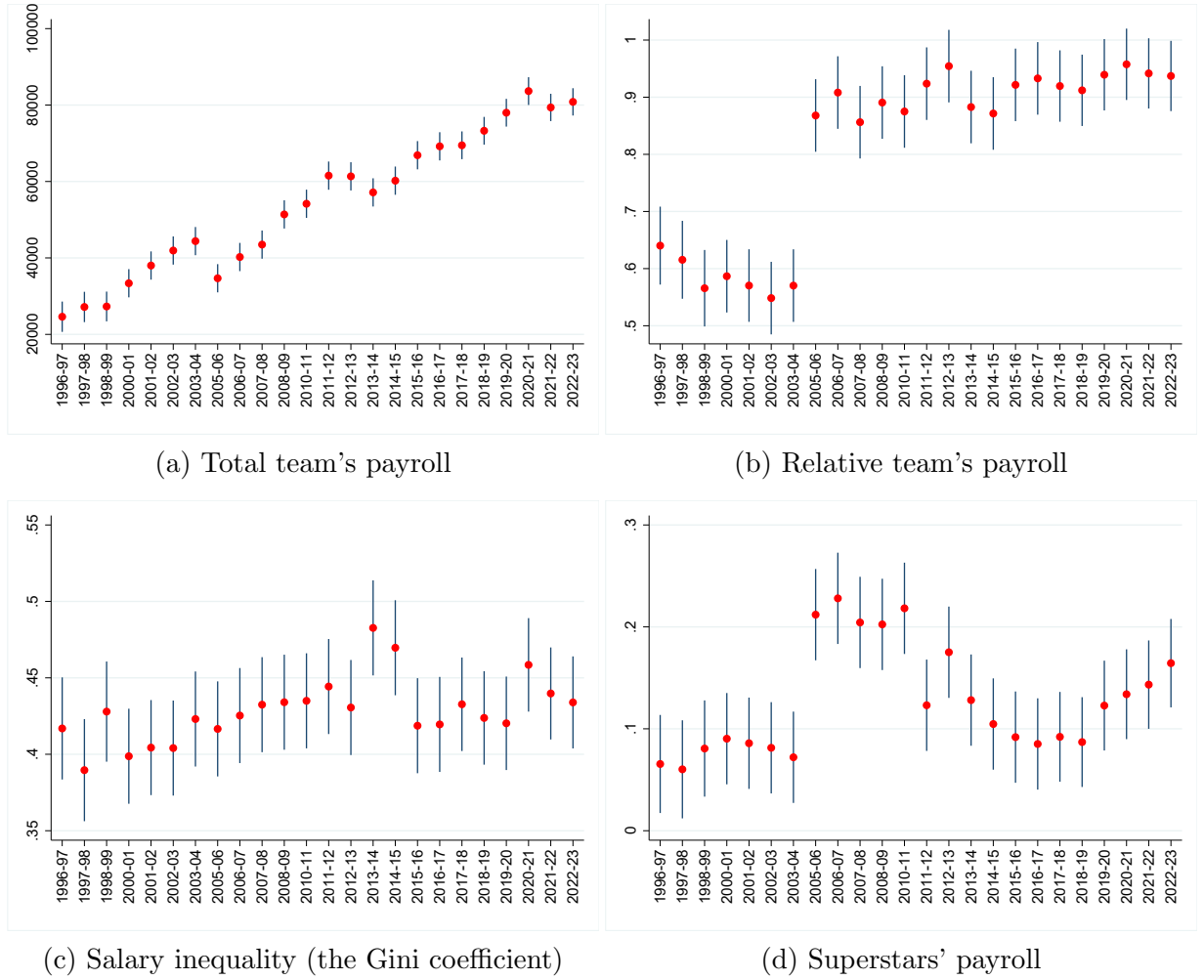
Season	Confirmed	% Change	Salary Cap	Salary Floor	Min. Salary
2025-26	NHL Estimate	5.14%	\$92,000,000	\$68,000,000	\$775,000
2024-25	NHL Estimate	5.03%	\$87,700,000	\$64,700,000	\$775,000
2023-24	Jun. 23, 2023	1.21%	\$83,500,000	\$61,700,000	\$775,000
2022-23	Mar. 29, 2022	1.23%	\$82,500,000	\$61,000,000	\$750,000
2021-22	Jul. 1, 2021	0.00%	\$81,500,000	\$60,200,000	\$750,000
2020-21	Jul. 10, 2020	0.00%	\$81,500,000	\$60,200,000	\$700,000
2019-20	Jun. 22, 2019	2.52%	\$81,500,000	\$60,200,000	\$700,000
2018-19	Jun. 21, 2018	6.00%	\$79,500,000	\$58,800,000	\$650,000
2017-18	Jun. 18, 2017	2.74%	\$75,000,000	\$55,400,000	\$650,000
2016-17	Jun. 21, 2016	2.24%	\$73,000,000	\$54,000,000	\$575,000
2015-16	Jun. 23, 2015	3.48%	\$71,400,000	\$52,800,000	\$575,000
2014-15	Jun. 27, 2014	7.31%	\$69,000,000	\$51,000,000	\$525,000
2013-14	Jan. 6, 2013	7.17%	\$64,300,000	\$47,500,000	\$525,000
2012-13	Jun. 28, 2012	-6.69%	\$60,000,000	\$44,000,000	\$525,000
2011-12	Jun. 23, 2011	8.25%	\$64,300,000	\$48,300,000	\$525,000
2010-11	Jun. 24, 2010	4.58%	\$59,400,000	\$43,400,000	\$500,000
2009-10	Jun. 26, 2009	0.18%	\$56,800,000	\$40,800,000	\$500,000
2008-09	Jun. 26, 2008	12.72%	\$56,700,000	\$40,700,000	\$475,000
2007-08	Jun. 29, 2007	14.32%	\$50,300,000	\$34,300,000	\$475,000
2006-07	Jun. 27, 2006	12.82%	\$44,000,000	\$28,000,000	\$450,000
2005-06	Jul. 22, 2005	0.00%	\$39,000,000	\$23,000,000	\$450,000

Notes: Data is taken from the website [CapFriendly](#).

During the regular season, the teams follow the rules regarding how they can form their rosters, and these rules cannot be violated. First, the number of officially registered players on the roster is 23. Second, there is a salary cap, so none of the teams can spend on salaries more than this predetermined amount of money. At the same time, there is a salary floor, so the teams cannot spend too little money. This was made to deprive teams of the opportunity to specifically take a low place in order to get the best positions in the draft of young players. Furthermore, it protects the players from understating their salaries and increases the competitiveness of the league. Finally, there are some rules regarding the minimum and maximum wages for each player. The maximum wage is usually 20% of the total salary cap, while the minimum wage is about \$750,000. All these rules are summarized in Table 1, which provides information on the salary cap for all years since its introduction.

As can be seen from Table 1, the salary cap was invented only in 2005 after negotiations between the league, teams, and players. This change affected the teams' decisions significantly, as represented in Figure 1. In this figure, I draw average levels of the over-all team's payrolls, relative payrolls (the payroll divided by the maximum payroll in the

Figure 1: Dynamics of the main teams' characteristics from 1996/97 to 2022/23



Notes: These figures represent the average values of one of the considered variables for each season in the dataset (red dots) and 95% confidence intervals (dark blue lines). Data is collected from the websites [Rodney Fort's Sports Economics](#) for seasons 1996/97–2008/09; [Spotrac.com](#) for seasons 2010/11–2022/23. Seasons 1999/00 and 2009/10 are absent due to a lack of data, while the season 2004/05 was not held due to lockout.

league in the considered season), inequality measure (the Gini coefficient), and superstars' payroll. Here, I assume that the superstars are those who earn more than 10% of the highest team's payroll in the league. Figure 1 suggests that after the introduction of the salary cap, the total expenditures of the average team dropped, but the relative expenditures increased. This may mean that the teams became closer in terms of how much they spend, so the way they distribute money among players (i.e., the inequality) starts to matter more. Speaking of the superstars' payroll, there is an increase after the introduction of the salary cap that is caused by how I define the superstars. Since more teams start spending closer to the maximum amount, there are more players getting more than 10% of the maximum payroll in the league.

Thus, the period after the introduction of a salary cap is particularly interesting for the analysis, as the teams became constrained in how much they spent and the distribution

of the salaries could become more important. These factors may be especially important in recent years, when the average relative payroll increased even more in comparison to the first seasons after the introduction of the salary cap. Hence, I focus on these seasons, which were not covered in the previous studies.

3 Theoretical Framework

In this section, I model the trade-off that arises when hockey teams operate under a hard salary cap and want to hire superstars. I start with the introduction and explanation of the main assumptions. Then I run the simulations for different sets of exogenous parameters and discuss the obtained results.

3.1 Main Assumptions

I develop a theoretical model to find under which conditions the teams might be interested in hiring superstars. For this purpose, I model the decision-making process of one team.

Let's consider a team that has to hire n players. First, I assume that the only difference between the hired players is their talent (q_i) and, therefore, the wage that they earn (w_i). Second, I assume that these wages are fair, so players of higher quality earn more. This assumption allows me to use the wages as a proxy for the player's talent. Moreover, as I model the demand side of the market for players, I assume that there is unlimited supply for the players of each talent, so the teams can just choose the players from the unlimited pool and pay them a fair wage. Thus, the team choice can be represented as the choice of the n -dimensional vector of players' wages, w .

Now let me describe exactly how this choice is made. The team has some objective function that can be interpreted as the team's productivity, and hence, it is maximized by the team. Let this function be denoted as $F(w)$. The team maximizes this function, subject to a few constraints. First, it is the budget constraint that follows from the rule about a hard salary cap. All teams must spend no more than some predetermined amount of money and cannot violate this rule. In my model, I normalize the cap to be equal to 1, so the wages of the players can be interpreted not as dollars but as the shares of the total team's cap. Then the budget constraint can be written in the following way:

$$\sum_{i=1}^n w_i \leq 1 \quad (1)$$

Other constraints that cannot be violated also follow from the National Hockey League rules. These are the constraints regarding the minimum and maximum wages. In each season, the NHL determines these wages, and all players must get salaries that lie in the

range between the minimum and maximum wages.² Therefore, the remaining constraints are:³

$$\forall i : w_i \in [w_{min}, w_{max}] \quad (2)$$

Speaking of the team's productivity function, I assume that it can be rewritten as a sum of the individual productivity functions of the players (f_i):

$$F(w) = \sum_{i=1}^n f_i(w) \quad (3)$$

Further, I assume that these individual performance functions depend on some player- and team-specific characteristics. First, the own talent of the player, which can be measured by his wage, should positively affect the player's performance. Second, I assume that inequality ($g(w)$) has a negative effect on the players. Third, I assume that the players benefit from having superstars among their peers, which I model by adding some superstar effect of the peers ($s_i(w)$). Finally, I allow for some heterogeneity among the players by introducing player-specific coefficients α_i and β_i that capture the size of inequality and superstar effects, respectively. Summing it all up, the individual performance functions are the following:

$$f_i(w) = w_i - \alpha_i g(w) + \beta_i s_i(w) \quad (4)$$

Now let me define $g(w)$ and $s_i(w)$ and explain why I have chosen these specific functions. First, $g(w)$ is some measure of inequality, so it should have the inverted U-shaped dependence on w_i for each i . One of the functions satisfying this condition is the Herfindahl-Hirschman index:

$$g(w) = \sum_{i=1}^n \left(\frac{w_i}{\sum_{i=1}^n w_i} \right)^2 \quad (5)$$

Second, $s_i(w)$ should be increasing in the number of superstars and the quality of them. Hence, I determine the superstar effect of the peers as a sum of wages spent on players that earn more than some exogenous superstar threshold, $\tilde{w}$:

$$s_i(w) = \sum_{j \neq i} w_j \cdot \mathbb{I}\{w_j > \tilde{w}\} \quad (6)$$

According to this function, if the player j earns more than a superstar threshold $\tilde{w}$, then he has a superstar effect on all his peers and has no additional effect on himself. Moreover,

²Actually, when the player signs his first contract, he signs the entry-level contract, and his salary may be lower than the minimum wage. But for simplicity, I do not take these cases into account.

³According to the rules of the NHL, there is also a salary floor constraint. It was invented to prevent teams from trading away their players to sacrifice their position in the league for a higher position in the following draft. Hence, this rule is not affecting the teams that maximize their performances, so I omit this constraint.

I assume that all superstar players are affected by other superstars identically, so:

$$\forall j \in \{w_j > \tilde{w}\} : \beta_j = \bar{\beta} \quad (7)$$

This additional assumption allows me to make the following denotations for simplification:

$$\alpha = \sum_i \alpha_i \quad \text{and} \quad \forall i \in \{w_i > \tilde{w}\} : \beta = \sum_{j \neq i} \beta_j \quad (8)$$

Thus, combining all the assumptions made above, the team maximizes the following function:

$$F(w) = \sum_i w_i (1 + \beta \cdot \mathbb{I}\{w_i > \tilde{w}\}) - \alpha \sum_{i=1}^n \left(\frac{w_i}{\sum_{i=1}^n w_i} \right)^2 \quad (9)$$

subject to constraints specified in (1) and (2).

3.2 Solution

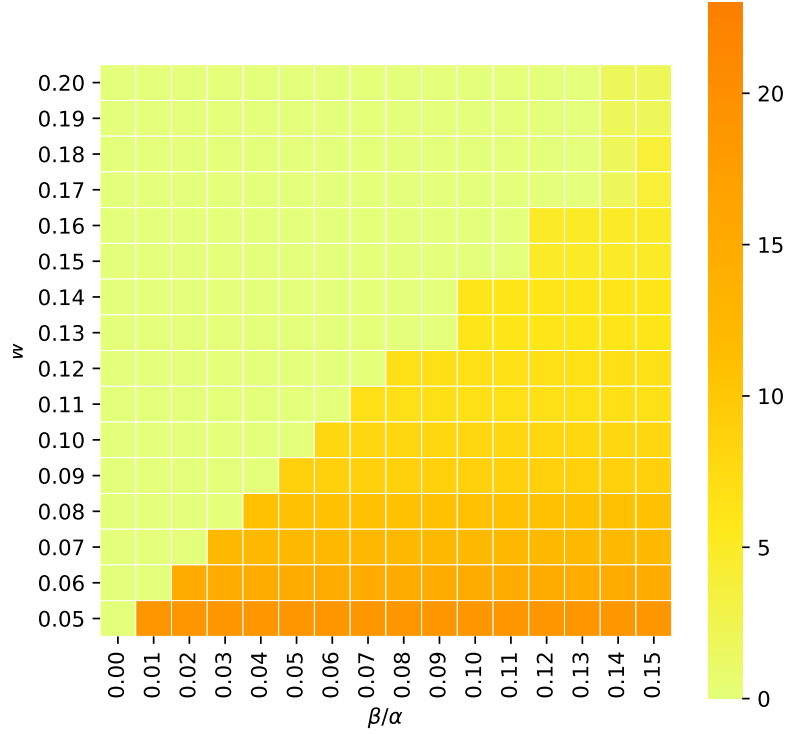
In this subsection, I present and discuss the solution to the maximization problem specified in the previous subsection.

The function that the team maximizes is not continuous, so there might be multiple local extrema. Furthermore, there are $n + 1$ constraints that allow for the opportunity of the corner solutions. Thus, derivation of the closed-form solution is very problematic, so I have written the script that finds the numerical solution for this problem for any given set of exogenous parameters. I describe the algorithm for finding the solution in Appendix A and now present the results of running the script for different parameters.

There are six exogenous parameters in my model: the minimum and maximum wages w_{min} and w_{max} , the number of players on the roster n , the threshold superstar salary $\tilde{w}$, the size of the inequality effect α , and the superstar effect β . w_{min} and w_{max} are chosen with respect to the official rules of the NHL that were described in Section 2. According to Table 1, in the 2022/2023 season, the salary cap was \$82,500,000, and the minimum wage was \$775,000, while the maximum amount should not have exceeded 20% of the team's salary cap. Hence, I assume that $w_{min} = \frac{775}{82500} \approx 0.009$ and $w_{max} = 0.2$. The roster size n is chosen to be equal to 23, which is in line with the NHL rules.

The remaining parameters $\tilde{w}$, α , and β cannot be chosen based on the rules of the NHL, so I find the optimal salary allocations for different values of these parameters. I present the results of these simulations in Figure 2. In the X-axis, there is the ratio of the superstar effect β to the inequality effect α . In Appendix A, I show that the size of one effect relative to another one matters but not the absolute values of the effects. This is the reason why I plot the ratio of two coefficients in the X-axis. In Figure 2, I allow this ratio to vary from 0 to 0.15, and the higher values of it correspond to the stronger superstar effect relative to the negative inequality effect. In the Y-axis, I plotted

Figure 2: Number of superstars in the optimal solution for given parameters



Notes: In the X-axis, I plot the superstar threshold $\tilde{w}$. In the Y-axis, I plot the ratio of the superstar effect to the inequality effect. The color of each square corresponds to the number of superstars hired with respect to the theoretical solution.

the superstar salary threshold, which varies from 0.05 to 0.2. The higher values of this parameter suggest that the players should be of higher quality so they can start having a positive effect on their peers. At the same time, having a superstar becomes costly with increases in $\tilde{w}$ because this parameter corresponds to the minimum superstar wage. I find the optimal distribution for different sets of parameters, and the colorbar demonstrates how many superstars should be hired for each of these sets.

Figure 2 suggests that for each threshold superstar salary $\tilde{w}$, there is some value of the ratio β/α when the optimal decision switches from having no superstars to hiring them. Furthermore, for the higher values of $\tilde{w}$, the superstar effect should be higher so that the team decides to hire superstars. If it is not high enough, the team chooses not to hire any superstars.

Additional results of the simulations are presented in Appendix A. For instance, Figure 4 suggests that if the team decides to hire superstars, it spends almost all its budget on them. The superstars' payroll decreases with $\tilde{w}$ but stays vast for even large values of $\tilde{w}$. Figure 5 demonstrates that with increases in $\tilde{w}$, the average salary paid to superstars increases as well. This means that when the superstars are quite cheap, the teams want to invest not in the quality of superstars (hire superstars with higher wages) but in the

number of superstars, which is explained by the high peer effects of each new superstar. Finally, the decision to have superstars makes the team cut the wages of other players, as seen in Figure 6. Until the team decides to hire superstars, all players get an equal amount of money, but when the team changes its mind, many players start earning close to the minimum wage.

These results are very intuitive. First, when the benefits from superstars (peer effects) start dominating the negative consequences of hiring them (high inequality), it becomes optimal to have superstars by cutting the wages for other players. But when the inequality effect is too high, it is better to have a team consisting of equal players in terms of their qualities and salaries. Furthermore, when the superstars become more expensive (higher $\tilde{w}$), then the team hires fewer superstars, and the positive effects of superstars must be much higher, so the team decides to hire at least one superstar. But if it is high enough, the team spends a significant portion of its budget on the superstars.

Thus, this model produces realistic results, and it can be used for the prediction of optimal decisions for real-world teams. However, to make any predictions about them, I need to make additional assumptions about the remaining exogenous parameters, and I will do that after the estimation of them in the following sections.

4 Data

In this section, I describe the data that I use for the estimation of the parameters of the model specified in the previous section.

4.1 Salaries

First, I collect the data on the players' salaries for seasons from 2011/12 to 2022/23 from the website [Spotrac.com](https://www.spotrac.com).

Using this data, I first calculate the total payrolls for all teams in all seasons. These payrolls are just the sum of all salaries paid to the players on the roster. However, in estimations, I use the relative payroll, which is the payroll divided by a salary cap in the considered season. This allows me to take into account the fact that salaries and caps in the NHL are growing. After this normalization, I control for these two patterns, which makes the observations from different seasons more comparable to each other.

Second, I calculate the inequality measures for all teams in all seasons. The preferred measure is the Herfindahl-Hirschman index, as it was used in the theoretical model, but I also use the Gini coefficient and the coefficient of variation (CV) for robustness checks.

Third, I calculate a few team-level characteristics required for the estimation of the superstar effect. They are the payroll of the superstars and the number of superstars. Speaking of how I defined the superstars, I tried a few different threshold values of salaries

(8%, 10%, and 12% of the league's cap) from which the player is considered to be a superstar. I will further discuss why I have chosen these values.

Fourth, since I have the data on the positions of all players, I divide them into attacking (wingers and centers) and defending players (goalies and defenders) and calculate payrolls, inequality measures, and superstars' measures for defenders and attackers separately.

Finally, as I focus not only on the team-level but also on player-level performances, I adjust the total payroll, the superstar's payroll, and the number of superstars so that they do not take into account the data on the considered player. In other words, I subtract player's salary from the total payroll and get net payroll. Moreover, if the player is a superstar, I subtract his salary from the superstars' payroll and decrease the number of superstars by one to take into account only the peer effects of the superstars.

4.2 Performances

The second source for the data is the website Hockey-Reference.com, from which I collect the data on team- and player-level performances in the regular season.

First, for the teams, I use the share of the maximum number of points that the team actually scored as the measure of productivity. The maximum possible number of points is 164 (which is the number of games in the regular season multiplied by two), so this performance measure is just the number of points divided by 164. For some of the specifications, I also use goals for and against per game to estimate the productivity of attacking and defending players.

Second, as I focus on player-level performances, I collect data on them as well. The main measure of performance that I use is the point share, which is the estimated number of points contributed by a player. [James and Henzler \(2002\)](#) developed a measure of players' contribution to the success of the team for baseball, and then their approach was adapted and used for many other kinds of sports. Hockey is not an exception. The Hockey-Reference.com website provides estimates of the NHL players' contributions that were calculated by the team developing this website. This measure takes into account both the offensive and defensive activities of all players, setting different weights for players in different positions. Thus, this measure should allow me to compare the players in different positions to each other, which makes it the most preferred measure for my analysis. Additionally, I collect the data on points (goals and assists) per game and total time on ice for all players for robustness checks.

4.3 Descriptive Statistics

Finally, I merge the data collected from these two sources and get the team-level and player-level datasets containing information on both productivity and salaries. Tables [2](#) and [3](#) provide the summary statistics of the player- and team-level data, respectively. In

Table 2: Descriptive statistics of player-level data

VARIABLES	(1) N	(2) Mean	(3) SD	(4) Min	(5) Max
Relative salary	8,107	0.0359	0.0307	0.00004	0.159
$D(w > 0.1)$	8,107	0.0417	0.200	0	1
$D(w > 0.12)$	8,107	0.0132	0.114	0	1
$D(w > 0.08)$	8,107	0.0984	0.298	0	1
Number of superstars ($\tilde{w} = 0.1$)	8,107	1.002	0.976	0	4
Superstars' payroll ($\tilde{w} = 0.1$)	8,107	0.116	0.116	0	0.489
Relative net payroll	8,107	0.902	0.100	0.485	1
Gini	8,107	0.441	0.0453	0.311	0.592
HHI	8,107	0.0639	0.00924	0.0446	0.108
CV	8,107	0.842	0.121	0.546	1.265
Points Share	8,107	2.858	2.802	-1.400	18.20
Time on ice (minutes)	8,107	889.0	520.9	1	2,412
Points (per game)	8,107	0.397	0.272	0	3
Points (per minute)	8,107	2.398	1.457	0	27.27

Notes: Data from [Spotrac.com](https://www.sportradar.com) for salaries and [Hockey-Reference.com](https://www.hockey-reference.com) for performance measures.

Appendix B, I also present Figures 7 and 8 that demonstrate the distributions of the main player- and team-level variables, respectively.

The first row of Table 2 provides the summary of the relative salaries. According to this table, the mean salary is 3.59% of the total salary cap, and there are no players getting more than 16%. The next three rows provide information on how many superstars are in my data. For example, if the superstar threshold is 10% of the salary cap, then there are 4.17% observations corresponding to the superstar players. It is not obvious which threshold value should be chosen, so I tried to choose the threshold for which there would be not too many and not too few superstars. I believe that 4.17% superstars corresponding to $\tilde{w} = 0.1$ is exactly what I needed. Hence, it is my preferred threshold value, which I use in most of the specifications, but I also try 8% and 12% as robustness checks. Using the preferred threshold, I find that an average team has one superstar, and about 11.6% of the salary cap is spent on him. Finally, I provide the summary statistics for the inequality and performance measures.

Table 3 shows the summary statistics for the similar variables, but at the team level. Additionally, I report the statistics for position-specific variables, such as the number and the payroll of superstars who are defenders and attackers, separately. Furthermore, the performance indicators are different from those that I use at the player level, as I discussed in the previous subsection.

Table 3: Descriptive statistics of team-level data

VARIABLES	(1) N	(2) Mean	(3) SD	(4) Min	(5) Max
Relative payroll	368	0.925	0.0940	0.596	1
Relative payroll (Attackers)	368	0.557	0.0966	0.294	0.846
Relative payroll (Defenders)	368	0.394	0.0743	0.141	0.699
Number of superstars ($\tilde{w} = 0.1$)	368	1.046	0.996	0	4
Number of superstars (Defenders)	368	0.321	0.512	0	2
Number of superstars (Attackers)	368	0.726	0.863	0	3
Superstars' payroll ($\tilde{w} = 0.1$)	368	0.121	0.118	0	0.489
Superstars' payroll (Attackers)	368	0.0852	0.105	0	0.412
Superstars' payroll (Defenders)	368	0.0360	0.0575	0	0.248
HHI	368	0.0644	0.00935	0.0446	0.108
HHI (Attackers)	368	0.112	0.0197	0.0733	0.169
HHI (Defenders)	368	0.149	0.0297	0.0912	0.332
Points (%)	368	0.558	0.0961	0.275	0.823
Goals For (per game)	368	2.844	0.369	1.830	4.110
Goals Against (per game)	368	2.844	0.378	1.890	4.090

Notes: Data from [Sportrac.com](https://www.sportradar.com) for salaries and [Hockey-Reference.com](https://www.hockey-reference.com) for performance measures.

5 Methodology

In this section, I explain the empirical model that I use for the estimation of inequality and the superstar effects.

I will start with the model that uses player-level data:

$$y_{ict} = \alpha + \beta_1 salary_{ict} + \beta_2 inequality_{ct} + \beta_3 superstars_{ict} + \delta_c + \delta_t + \epsilon_{ict} \quad (10)$$

On the left-hand side of the equation, I have y_{ict} , which is some measure of the productivity of the player i in season t playing for the club c . The variables that I use as productivity measures are described in the previous section. On the right-hand side, $salary_{ict}$ is the salary of a player divided by the league's cap. The variable $inequality_{ct}$ is the measure of inequality of wages in team c in season t (i.e., HHI, Gini, or CV). Another variable is $superstars_{ict}$, which is a total payroll of superstars of team c in season t , excluding player i if he is a superstar. Finally, δ_c and δ_t are team- and season-specific fixed effects, and ϵ_{ict} is an error term. Such a model estimates the coefficients of the equation estimated in (4) but also includes the season- and team-fixed effects.

If the salary is a good proxy for talent, then β_1 should be positive. The negative coefficient β_2 would mean that the adverse inequality effect has been preserved in recent seasons. Finally, if the superstars have positive peer effects on their teammates, β_3 is expected to be positive. These are the signs of the coefficient that I expect to observe in

the estimations.

It is worth noting that the variables corresponding to inequality and superstars are likely to be endogenous since they are the results of the team’s decisions. However, my identification strategy relies on the assumption that most of the omitted factors are either time- or team-invariant, and, hence, they are captured when I use the TWFE model. Nevertheless, this assumption might be violated if there are some omitted factors that are correlated with either inequality or superstars’ payroll. Unfortunately, this cannot be tested, but to increase confidence in the results, I add some additional control variables, like the payrolls of players other than superstars, to some of the specifications. Besides, I also use the number of superstars instead of their total payroll as an alternative measure of the superstars’ effect. In some specifications, I include both the superstars’ payroll and the number of superstars to study which of these factors is more important for the performance of the peers.

Furthermore, I study whether the effects are similar for players in different positions. For this purpose, I split the dataset into defenders and attackers and run separate regressions for them. In these regressions, I include the position-specific characteristics of the teams, so instead of one measure of inequality for the whole team, I include the separate measures for defenders and attackers. The same is done with the variables for measuring the superstar effect and net payrolls.

Finally, I run the regressions using the team-level data. In this case, I estimate the coefficients from the equation (9) from the theoretical part. In regression form, it can be written in the following way:

$$y_{ct} = \alpha + \beta_1 payroll_{ct} + \beta_2 inequality_{ct} + \beta_3 superstars_{ct} + \delta_c + \delta_t + \epsilon_{ict} \quad (11)$$

This regression is very similar to the one that I run using player-level data but differs in a few directions. First, I use team-level measures of productivity. Second, I include the team’s relative payroll $payroll_{ct}$ instead of the relative salary of the player. Third, variables for measuring the superstar effect include all superstars on the team. As this model is the closest to the maximized function from the equation (9), I use the coefficients from this specification to make predictions about the optimal number of superstars.

6 Results

In this section, I provide and discuss the results of the estimations. I start with the main results and discuss how they relate to the theoretical framework. Then I study the potential heterogeneity of the effects and discuss how it may affect teams’ decisions. Finally, I provide a few robustness checks for my results.

Table 4: Analysis of the superstar and inequality effects using the player-level data

VARIABLES	(1) Points Share	(2) Points Share	(3) Points Share	(4) Points Share	(5) Points Share
Relative salary	52.12*** (1.550)	52.11*** (1.549)	52.13*** (1.545)	51.24*** (1.576)	51.24*** (1.579)
HHI	-18.51*** (4.323)	-18.07*** (4.305)	-18.61*** (4.321)	-15.74** (6.756)	-15.67** (6.713)
Superstars' payroll	0.897*** (0.322)		1.144 (2.468)	1.083*** (0.377)	1.006 (2.434)
Number of superstars		0.102** (0.0381)	-0.0287 (0.293)		0.00911 (0.291)
D(Relative salary > 0.1)				0.304 (0.302)	0.304 (0.301)
Non-superstars' payroll				0.335 (0.508)	0.338 (0.509)
Constant	2.063*** (0.274)	2.038*** (0.273)	2.069*** (0.278)	1.614* (0.820)	1.608* (0.822)
Team FE	Yes	Yes	Yes	Yes	Yes
Season FE	Yes	Yes	Yes	Yes	Yes
Observations	8,107	8,107	8,107	8,107	8,107
Number of teams	32	32	32	32	32
R-squared	0.364	0.363	0.364	0.364	0.364

Notes: Estimated coefficients are obtained using the TWFE model and data on all available players except goalies for seasons from 2011/12 to 2022/23. Data is collected from [Spotrac.com](https://www.spotrac.com) for salaries and [Hockey-Reference.com](https://www.hockey-reference.com) for performance measures. Salaries and payrolls are divided by the league's cap. Superstars are assumed to be the players who get salaries higher than 10% of the league's cap. The payrolls of superstars and non-superstars do not include the salary of considered individual. Standard errors clustered at the team level are reported in parentheses. The p-values are denoted as follows: *** p<0.01, ** p<0.05, * p<0.1.

6.1 Main Results

First, I estimate the regression specified in (11). The results of this estimation are provided in Table 4. Column (1) shows the results of the regression discussed in the previous section, for which the dependent variable is the Points Share. Column (2) uses the number of superstars instead of their payroll, while the specification in Column (3) includes both of these variables. Columns (4) and (5) include the payroll of the players other than superstars and the dummy for whether the player is a superstar himself. The only difference between these two specifications is that the first one does not include the number of superstars as one of the control variables.

According to all columns in Table 4, the relative salary is always significant and positive, which means that wages are strongly correlated with performance. The coefficients corresponding to the inequality measure, which is the HHI in these regressions, are always negative and significant, which signals the presence of the negative inequality effect on players' performance. Speaking of the superstar effect, this table suggests that it is positive and significant. The coefficients corresponding to the measures of the superstar effect

are always significant when I use only one measure. If I use both the superstars' payroll and the number of superstars, both of the coefficients become insignificant, however, their magnitude does not differ too much. In other words, just the standard errors of the estimated coefficients increase what may be caused by the large correlation between two measures of the superstar effect. Hence, I tested the hypothesis for their joint hypothesis that both of the coefficients are zeros, and it was rejected at the 5% level with an F-test equal to 4.24. Finally, the remaining variables, such as the payroll of players other than superstars and the dummy for whether a person is a superstar, are insignificant, so there is no evidence for their impact on the players' performance.

Speaking of the magnitude, I refer to Table 9 from Appendix C. In this table, I used the logarithms of some variables, so the interpretation of the coefficients is easier. According to column (2) of Table 9, hiring one additional superstar increases the performance by 3.27 percentage points. However, 1% increase in inequality is associated with a 0.255% decrease in the player's productivity. Thus, the total effect of superstars is not obvious. To better understand which effect is stronger, let's consider an example.

Assume the team equally distributes its money among players. Then the HHI for this team is $\frac{1}{23} \approx 0.043$. But if it decides to hire one superstar with a salary equal to 10% of the league's cap and distribute the remaining money equally among other players, then the new HHI is $\frac{0.9^2}{22} + 0.1^2 \approx 0.047$. Therefore, hiring one superstar for this team increases the inequality by 9.3%, which corresponds to 2.37 percentage points decrease in productivity. Hence, with the direct effect of superstars equal to 3.27 percentage points, the total effect for the considered team appears to be only 0.9 percentage points, but it remains positive.

Nevertheless, this is just an example, but is hiring superstars optimal for an average team, and how many superstars should the team hire? To answer this question, I estimate the inequality and superstar effects using the team-level data, and the results of these estimations are provided in Table 5. In column (1), I present the results from the estimation of the specification, which corresponds to the objective function of the teams in the theoretical framework. Column (2) uses the number of superstars instead of their payroll, while column (3) includes both of these variables. In columns (4) and (5), I replace the relative total payroll with the payroll of non-superstar players. These specifications differ in only one way: I control for the number of superstars in (5), while I do not do this in (4).

The results of these estimations are the following. There is a negative and significant effect of inequality and a positive and significant effect of superstars. When I include both the superstars' payroll and the number of superstars, both coefficients are insignificant, but they are jointly significant. The effects of relative payroll and the non-superstars' payroll are positive and significant in all specifications in which I include them, but they do not strongly affect estimates of the superstar and inequality effects.

Table 5: Analysis of the superstar and inequality effects using the team-level data

VARIABLES	(1) Points (%)	(2) Points (%)	(3) Points (%)	(4) Points (%)	(5) Points (%)
Relative payroll	0.259*** (0.0760)	0.269*** (0.0769)	0.256*** (0.0754)		
HHI	-3.550*** (0.907)	-3.360*** (0.900)	-3.628*** (0.916)	-2.719*** (0.832)	-2.686*** (0.849)
Superstars' payroll	0.210*** (0.0686)		0.345 (0.482)	0.423*** (0.0658)	0.384 (0.440)
Number of superstars		0.0232*** (0.00754)	-0.0155 (0.0536)		0.00459 (0.0496)
Non-superstars' payroll				0.269*** (0.0636)	0.270*** (0.0639)
Constant	0.522*** (0.118)	0.501*** (0.119)	0.529*** (0.117)	0.459*** (0.102)	0.455*** (0.103)
Team FE	Yes	Yes	Yes	Yes	Yes
Season FE	Yes	Yes	Yes	Yes	Yes
Observations	368	368	368	368	368
Number of teams	32	32	32	32	32
R-squared	0.471	0.469	0.471	0.480	0.480

Notes: Estimated coefficients are obtained using the TWFE model and data on all teams for seasons from 2011/12 to 2022/23. Data is collected from [Sportrac.com](https://www.sportrac.com) for salaries and [Hockey-Reference.com](https://www.hockey-reference.com) for performance measures. Payrolls are divided by the league's cap. Superstars are assumed to be the players who get salaries higher than 10% of the league's cap. Standard errors clustered at the team level are reported in parentheses. The p-values are denoted as follows: *** p<0.01, ** p<0.05, * p<0.1.

Thus, all the specifications considered suggest that there is negative inequality and positive superstar effects. Based on the preferred specification, which is the one in column (1) of Table 5, the coefficients of the theoretical model are the following: $\alpha = 3.55$ and $\beta = 0.21$. As I mentioned when I described the data, I chose $\tilde{w}$ to be equal to 0.1 so that there are not too many and not too few superstars. Using these parameters, I get that the theoretical optimum is to hire 8 superstars, as can be seen from Figure 2 as $\tilde{w} = 0.1$ and $\beta/\alpha = 0.059$. Furthermore, for other specifications, the ratio of the superstar effect to the inequality effect is even higher, so the optimal decision for these sets of parameters is still hiring 8 superstars.

Therefore, my theoretical model using the parameters estimated using real-world data suggests that the teams should optimally hire eight superstars. But are there any teams that act this way, and how well do they perform? I try to answer these questions by plotting Figure 3. In this figure, I show how many superstars the teams hire and how well they perform in terms of the share of points scored. I find that there are no teams following the theoretical solution, as the maximum number of superstars per team is four. However, I believe that it might happen due to the scarcity of superstars. In the theoretical framework, I model the demand side of the market for superstar players. I also assume that the supply of players for each talent is unlimited. This is a pretty strong assumption, so its violation might be the reason why my model overestimates the number of optimally hired superstars. There is additional evidence in favor of this idea, as I find a positive correlation between the team's productivity and the number of superstars. Hence, probably the productivity of the teams continues increasing with the number of superstars up to 8, but we simply cannot observe it because of the absence of such observations.

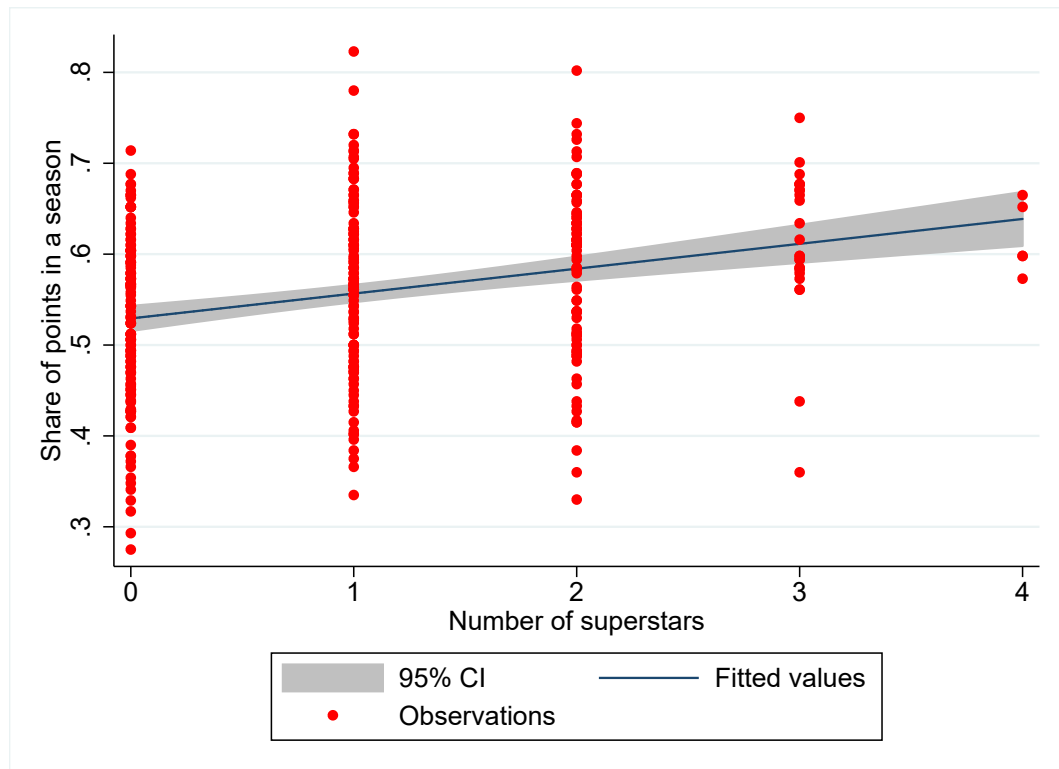
Summing it all up, I find some evidence in favor of negative inequality and positive superstar effects. This allows me to make predictions based on the theoretical framework discussed earlier. However, this framework overestimates the number of optimally hired superstars, which may be caused by not taking into account the limited supply of superstar players.

6.2 Heterogeneity of the Effects

In this subsection, I study whether the inequality and superstar effects are similar for all players. Based on the results of these estimations, I draw additional conclusions about the optimal choices regarding the number of superstars.

To begin with, I study whether the results differ by players' talent. In other words, do more professional players react to inequality and superstars differently when they are compared to the less professional ones? I assume that the professionalism of the player is determined by his salary. Then Table 6 presents the results of the study of this heterogeneity. In column (1), I use the data on the players that earn less than the median

Figure 3: Number of superstars and team's performance



Notes: This figure uses data on all available players except goalies for seasons from 2011/12 to 2022/23. Data is collected from [Spotrac.com](https://www.spotrac.com) for salaries and [Hockey-Reference.com](https://www.hockey-reference.com) for performance measures. In the X-axis, I plot the number of superstars in the team assuming that the superstars are players getting salaries higher than 10% of the league's cap. In the Y-axis, I plot the share of points that the teams scored in the season. Red dots correspond to the team-level observations, dark blue line shows the simple linear dependence of the performance on the number of superstars, and grey are stands for 95% confidence interval of the estimate.

Table 6: Analysis of the effects for players of different quality

VARIABLES	(1) Points Share $w < 0.022$	(2) Points Share $w \geq 0.022$	(3) Points Share Top 5	(4) Points Share Top 3	(5) Points Share Superstars
Relative salary	98.33*** (9.157)	56.10*** (2.854)	51.23*** (3.644)	51.62*** (5.105)	47.34*** (16.45)
Superstars' payroll	0.288 (0.364)	2.156*** (0.699)	1.873** (0.718)	2.377** (0.976)	3.324 (3.629)
HHI	-20.56** (7.666)	-10.19 (11.27)	-4.598 (11.80)	-6.573 (16.31)	16.27 (54.78)
Non-superstars' payroll	-0.722 (0.515)	1.631* (0.876)	2.471** (1.001)	3.295** (1.297)	8.258* (4.556)
Constant	2.371** (0.904)	-0.188 (1.416)	-0.741 (1.545)	-1.261 (1.976)	-6.098 (7.150)
Team FE	Yes	Yes	Yes	Yes	Yes
Season FE	Yes	Yes	Yes	Yes	Yes
Observations	4,061	4,046	3,033	1,826	338
Number of teams	32	32	32	32	30
R-squared	0.083	0.306	0.283	0.269	0.402

Notes: Estimated coefficients are obtained using the TWFE model and data on all available players except goalies for seasons from 2011/12 to 2022/23. Data is collected from [Spotrac.com](https://www.spotrac.com) for salaries and [Hockey-Reference.com](https://www.hockey-reference.com) for performance measures. Salaries and payrolls are divided by the league's cap. Superstars are assumed to be the players who get salaries higher than 10% of the league's cap. The payrolls of superstars and non-superstars do not include the salary of considered individual. Standard errors clustered at the team level are reported in parentheses. The p-values are denoted as follows: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

wage in the dataset. In this regression, I include the relative salary, the HHI, superstars' payroll, and non-superstars' payroll. Column (2) utilizes the same specification, but it uses the data on the players earning above the median wage. Column (3) presents the results of the same specification, but I consider only the players who are among the top 5 players in their clubs in their positions. For column (4), I consider only the players who are in the top 3 of their team. Finally, I run the regressions using the data only on the superstars in column (5).

The results presented in Table 6 suggest that the highest negative effect of inequality is observed for the players earning below the median wage. For all other specifications, the magnitude of the effect is lower, and it is insignificant. Speaking of the superstar peer effects, it seems that only top players benefit from them. The coefficient is close to 0 and insignificant in the first column, but it is much higher and significant in all other specifications except the latest one. In the latest specification, the coefficient is even higher than in the previous ones, but it is insignificant, probably because of the lower number of observations. Finally, spending on players other than superstars increases the performance of top players as well as the payrolls of superstars. Thus, these results suggest that having superstars and strong peers is beneficial for the top players, but the players

below the median level do not benefit from it.

Another source of potential heterogeneity is the different effects on players with different roles. For example, defenders might benefit more if their team has good forwards instead of other defenders. Hence, I try to test this and present the results in Table 7. In column (1), I just use the data for all attacking players and regress Points Share on the player's salary, payroll of superstar defenders, payroll of superstar attackers, and the HHI calculated using all players. In column (2), I include separate inequality indexes for the defenders and attackers instead of the common index. In column (3), I also include the spending on non-superstar defenders and attackers as the control variables. In column (4), I estimate the same specification but use only the data on the top 5 attackers from each team. Finally, columns (5)–(8) present the results of similar specifications, but I use the data on the defenders in these regressions.

According to Table 7, I get the following results. First, speaking of the attackers, they benefit from having superstar defenders, while the effect of superstar forwards is negative and insignificant. In other words, superstars who are playing in another position are complementary for the players. However, the superstars who have the same role as the player either have no effect or have a negative effect on the player's performance. This can be caused by the competition among these players and superstars. At the same time, the inequality effect is usually negative but often insignificant. Hence, probably, attackers are not affected by inequality as strongly as defenders. Furthermore, the specifications, including non-superstar payrolls, suggest that higher attackers' payrolls are associated with the worse performance of attackers, while increases in defenders' payrolls increase attackers' performance.

For defenders, I observe very similar results. They benefit from having good attackers, but superstar defenders have a small and insignificant effect. The same is true for spending on non-superstar players: spending on other defenders decreases their performance, while expenditures on attackers increase productivity. Speaking of the inequality indicators, I find that the coefficients corresponding to them are always negative, but they are significant only for HHI computed for attackers. In other words, defenders perform worse if there is higher inequality among attackers. This could be interpreted as high inequality among attackers leading to low chemistry among them, and their complementary power to the actions of defenders is not as strong as it could be.

Thus, both for defenders and attackers, I find that they contribute less to the success of their team when there are many good players with the same role. In other words, competition may worsen the players' performance. But what exactly drives this result? Do players get less time on the ice and, therefore, contribute less? Or are there any psychological factors that make the players perform worse?

Table 8 presents the results that help me answer these questions. Columns (1) and (5) of this table are the same as columns (1) and (5) of Table 7, respectively. Other columns

Table 7: Analysis of the effects for players of different positions

VARIABLES	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Points Share Attackers	Points Share Attackers	Points Share Attackers	Points Share Attackers (Top 5)	Points Share Defenders	Points Share Defenders	Points Share Defenders	Points Share Defenders (Top 5)
Relative salary	53.11*** (1.969)	52.90*** (1.952)	50.87*** (2.239)	54.10*** (4.771)	49.69*** (2.335)	49.89*** (2.337)	46.87*** (3.039)	49.98*** (4.401)
HHI	-14.02*** (4.839)				-25.81*** (5.217)			
HHI (A)		2.525 (2.634)	-3.603 (3.105)	-5.980 (7.850)		-14.69*** (2.605)	-7.738* (4.028)	-10.92** (4.675)
HHI (D)		-7.050*** (1.371)	-1.460 (1.845)	5.103 (4.909)		-0.421 (1.727)	-3.838 (2.839)	-6.178 (4.243)
Superstars' payroll (A)	0.337 (0.448)	-0.345 (0.440)	-0.715 (0.463)	-0.117 (1.264)	2.093*** (0.539)	2.539*** (0.541)	3.056*** (0.568)	3.446*** (0.805)
Superstars' payroll (D)	1.328* (0.765)	1.801** (0.728)	2.978*** (0.753)	4.133* (2.288)	-0.843 (1.188)	-1.288 (1.140)	-1.869 (1.211)	-2.024 (2.106)
Non-superstars' payroll (A)			-1.422** (0.573)	-1.605 (1.402)			1.721** (0.752)	2.861*** (0.986)
Non-superstars' payroll (D)			3.138*** (0.694)	7.102*** (2.135)			-1.906* (0.988)	-1.326 (1.981)
Constant	1.584*** (0.320)	1.496*** (0.349)	0.865 (0.810)	-1.525 (1.683)	2.909*** (0.351)	2.933*** (0.337)	2.484** (0.946)	2.244 (1.498)
Team FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Season FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	5,223	5,223	5,223	1,675	2,884	2,884	2,884	1,358
Number of teams	32	32	32	32	32	32	32	32
R-squared	0.386	0.389	0.392	0.316	0.344	0.346	0.348	0.324

Notes: Estimated coefficients are obtained using the TWFE model and data on all available players except goalies for seasons from 2011/12 to 2022/23. Data is collected from [Spotrac.com](https://www.spotrac.com) for salaries and [Hockey-Reference.com](https://www.hockey-reference.com) for performance measures. Salaries and payrolls are divided by the league's cap. Superstars are assumed to be the players who get salaries higher than 10% of the league's cap. The payrolls of superstars and non-superstars do not include the salary of considered individual. "A" in the parentheses stands for Attackers, while "D" is for Defenders. Standard errors clustered at the team level are reported in parentheses. The p-values are denoted as follows: *** p<0.01, ** p<0.05, * p<0.1.

in Table 8 use the same specification but another dependent variable. In columns (2) and (6), I use the total time on ice instead of the Points Share, while in (3) and (7), I use the number of goals and assists per game. In columns (4) and (8), I use the average number of goals and assists per 100 minutes on ice as the measure of productivity.

According to Table 8, both attackers and defenders start having less time on ice when there are many good players in the same position. At the same time, the points per game and per minute are negatively affected for attackers by the superstar and non-superstar payrolls of other attackers. For defenders, I find no effect of the payrolls on this measure of productivity, which is not surprising as defenders are not expected to score a lot. Thus, the results for attackers suggest that there is some evidence that the higher competition decreases both time on ice for the players and their performance.

Summing it all up, there is some heterogeneity among the players. First, top players are less vulnerable to the negative effects of inequality and benefit more from the presence of superstars. Second, superstar attackers have a complementary effect on defenders, and superstar defenders boost attackers' performance. However, the peer effect of superstars on players with the same role as theirs may be negative because of the increased competition. This competition both decreases the players' time on ice and their productivity, which is measured as the number of scored points divided by time on ice.

6.3 Robustness Checks

To make the obtained results more credible, I also provide some robustness checks. For this purpose, I try alternative functional forms, some other inequality measures, different threshold values of superstars, and other productivity measures. Furthermore, I focus only on the top teams and check whether the results are the same for the top and average teams.

To begin with, in Table 9 in Appendix C, I try some other functional forms of some variables. For instance, I use the logarithm of Points Share in all the specifications and the logarithm of HHI in the first five columns. In columns (4) and (5), I assume that the superstar effect may be non-linear, so I include the squares of different measures for the superstar's effect. In column (6), I include the squared HHI to check whether it is significant or not.

According to Table 9, the results are similar in all alternative specifications. The superstar and inequality effects remain significant in the specifications with logarithms. At the same time, specifications with squared variables suggest there is not enough evidence for the quadratic dependence on any of the variables.

Additionally, I use some alternative measures of inequality, such as the Gini coefficient and the coefficient of variation (CV). The results of the analysis at the player level are presented in Table 10 in Appendix C. According to this table, the coefficients for all

Table 8: Analysis of the effects for players of different positions (alternative performance measures)

VARIABLES	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Points Share Attackers	Time on Ice Attackers	Points per Game Attackers	Points per 100 Minutes Attackers	Points Share Defenders	Time on Ice Defenders	Points per Game Defenders	Points per 100 Minutes Defenders
Relative salary	50.87*** (2.239)	6.995*** (318.0)	5.325*** (0.250)	20.38*** (1.163)	46.87*** (3.039)	9.151*** (473.4)	3.209*** (0.198)	9.758*** (0.850)
HHI (A)	-3.603 (3.105)	-473.2 (396.4)	-0.619* (0.321)	-3.726* (2.190)	-7.738* (4.028)	-295.6 (590.8)	-0.244 (0.292)	-0.440 (1.703)
HHI (D)	-1.460 (1.845)	178.1 (218.9)	0.0886 (0.187)	0.293 (1.359)	-3.838 (2.839)	-301.0 (425.0)	-0.209 (0.183)	-1.259 (0.918)
Superstars' payroll (A)	-0.715 (0.463)	-228.1*** (78.42)	-0.144*** (0.0419)	-0.506* (0.272)	3.056*** (0.568)	193.2** (90.75)	0.0488 (0.0463)	0.200 (0.254)
Superstars' payroll (D)	2.978*** (0.753)	224.5** (97.40)	0.146* (0.0768)	0.937* (0.476)	-1.869 (1.211)	-813.3*** (151.9)	-0.142 (0.110)	0.167 (0.623)
Non-superstars' payroll (A)	-1.422** (0.573)	-451.3*** (67.96)	-0.260*** (0.0576)	-1.067*** (0.355)	1.721** (0.752)	160.3 (117.9)	0.0264 (0.0472)	0.261 (0.300)
Non-superstars' payroll (D)	3.138*** (0.694)	218.0** (99.22)	0.226*** (0.0657)	1.357*** (0.397)	-1.906* (0.988)	-1,137*** (138.0)	-0.168*** (0.0513)	-0.327 (0.263)
Constant	0.865 (0.810)	713.3*** (104.0)	0.346*** (0.0735)	2.471*** (0.461)	2.484** (0.946)	1,027*** (149.6)	0.280*** (0.0597)	1.361*** (0.353)
Team FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Season FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	5,223	5,223	5,223	5,223	2,884	2,884	2,884	2,884
Number of teams	32	32	32	32	32	32	32	32
R-squared	0.392	0.362	0.422	0.264	0.348	0.385	0.257	0.123

Notes: Estimated coefficients are obtained using the TWFE model and data on all available players except goalies for seasons from 2011/12 to 2022/23. Data is collected from [Sportrac.com](https://www.sportrac.com) for salaries and [Hockey-Reference.com](https://www.hockey-reference.com) for performance measures. Salaries and payrolls are divided by the league's cap. Superstars are assumed to be the players who get salaries higher than 10% of the league's cap. The payrolls of superstars and non-superstars do not include the salary of considered individual. "A" in the parentheses stands for Attackers, while "D" is for Defenders. Standard errors clustered at the team level are reported in parentheses. The p-values are denoted as follows: *** p<0.01, ** p<0.05, * p<0.1.

alternative inequality measures are negative and significant. Furthermore, the superstar peer effect remains significant in all the specifications.

I also run regressions using the data at the team level. The results of these regressions are presented in Table 11 in Appendix C. This table suggests that the inequality effect is negative for all measures of inequality, but for the Gini coefficient and CV, it is insignificant. One way to interpret these results is the following. In the previous subsection, I got that the inequality effect for the top players is very small and even insignificant. Hence, as these top players are usually the ones who contribute the most to the success of their team, it could be the case that the effect of inequality on the team's result is not very high. Thus, the insignificant results in Table 11 may be driven by the fact that new measures of inequality pay less attention to the non-top players in comparison to the HHI. Nevertheless, even if we consider that the inequality effect is insignificant, it makes hiring superstars even more profitable, so the theoretical prediction for other measures of inequality remains the same.

Then I use some alternative definitions of superstars and present how they change the main results. For the main part of my analysis, I used the 10% threshold for the superstars. In other words, all players earning more than 10% of the league's salary cap were defined as superstars, while all others were non-superstars. I used this threshold because I wanted to have not too many and not too few superstars, and I believe I achieved this goal as there were 4.17% players who were superstars. But now I try some alternative threshold levels, i.e., 8% and 12%.

The results of estimations of the baseline models on the player-level for alternative definitions of superstars are presented in Table 12 in Appendix C. According to this table, the effect of inequality remains negative and significant in all specifications. At the same time, the effect of superstars decreases and becomes insignificant. Probably, these results are driven by the fact that alternative definitions of superstars are worse, and they either capture fewer superstars (in the case of $\tilde{w} = 0.12$) or more non-superstar players (in the case of $\tilde{w} = 0.08$).

In Table 13 in Appendix C, I present the results of specifications using the team-level data. As was the case with player-level data, the inequality coefficients do not change much, but the superstar effects decrease and become insignificant in some of the specifications. Again, I attribute it to the fact that 12%-threshold is too harsh and does not take into account all superstars, while 8%-threshold is, contrary, too soft, which leads to taking into account the non-superstar players that may bias the estimates.

After that, I study how the results change if I use some alternative measures of productivity. Speaking of the player level, I partially did it when I focused on the heterogeneity among players having different roles in Table 8. Therefore, now I focus on the team-level data. Using it, I run the regressions similar to the ones in Table 8 and present them in Table 14 in Appendix C.

First, according to this table, inequality among both attackers and defenders has a negative effect on the share of points that the team scores. At the same time, the relative payrolls of attackers and defenders increase this measure of productivity as well. Finally, superstar forwards also have a positive effect, while the effect of superstar defenders is insignificant. Then I run the same regressions, but the dependent variable is the number of scored goals per game. For this performance measure, I find that the effect of inequality is negative but insignificant, the superstars do not have a significant effect, and there are more benefits from increasing the payrolls of defenders than attackers. This result is consistent with what I got earlier at the player level. According to Table 7, attackers benefit more from having good defenders, but having other good attackers may decrease their performance through increased competition. Therefore, hiring good forwards increases the team's scoring potential since it has more professionals among attackers. However, these players compete with each other, which slightly decreases the benefits of having many good attackers. Anyway, the overall effect measured in columns (3) and (4) of Table 14 is significant. Similar patterns are observed when I use the goals conceded as a measure of the team's productivity. For instance, expenditures both on attackers and defenders decrease the number of conceded goals, but the effect of attackers is higher in absolute values. It can happen because of the competition among defenders, which slightly decreases the benefits of having good defenders. Speaking of the inequality measure, the coefficients for it are positive and even significant for defenders. The positive coefficient in these regressions means that this variable increases the number of conceded goals, which signals a bad team's productivity. Thus, these specifications suggest that inequality among defenders decreases their productivity.

Finally, I analyze whether the baseline results are different for teams spending close to the league's cap and other teams. For this purpose, I provide Table 15 in Appendix C using the team-level data. In this table, I use all the available data in columns (1) and (2). In columns (3) and (4), I leave only the observations corresponding to the teams that spend more than 90% of the league's cap, while the regressions in columns (5) and (6) are run using the observations with payrolls higher than 95% of the league's salary cap.

Table 15 suggests that there is a positive superstar effect for both the top and non-top teams in terms of total payrolls, so all teams benefit from hiring the superstars. Speaking of the inequality effect, it is negative and significant, but for the top teams it is slightly lower, so, probably, these teams better overcome the potential issues arising from high inequality.

Summing it all up, I tried alternative measures of inequality and productivity, alternative definitions of superstars, alternative functional forms of regressions, and split the dataset by the relative payroll. All in all, the signs of the effects stayed the same as they were in the baseline, but sometimes the magnitude decreased and made the coefficients insignificant. In terms of inequality, I attribute it to the fact that alternative measures

pay less attention to the non-top players. Speaking of definitions of superstars, I think that some of the results became insignificant because of the less precise definition of superstars. Nevertheless, the general results about the positive superstar effect and the negative inequality effect seem robust.

7 Conclusion

In my thesis, I investigate the optimality of hiring superstars in the National Hockey League. I find the positive effect of superstars on their peers' and, therefore, the team's performance. At the same time, the inequality produced by hiring superstars decreases these positive effects. Nonetheless, after estimating the parameters and substituting them into my theoretical model, I get that teams should hire superstars if they act optimally.

Further, addressing the potential heterogeneity of the effects, I find that top players benefit more from the superstars among their peers. Moreover, the players are affected differently by players in different positions. Specifically, the peer effect is positive if the player and superstar's efforts are complementary, but it is negligible and sometimes even negative if they have the same role.

Taking these results seriously, the NHL teams' managers should hire the superstars if they have money for them and if there are available superstars on the market. However, it is not wise to invest all the money in the superstars who will compete with each other. It would be more beneficial to equally distribute the superstars among different positions. Thus, returning to the example of the Toronto Maple Leafs, they should probably not abandon the concept of hiring superstars. Nevertheless, they may perform better if they hire a few superstar defenders instead of investing all their money in attackers.

It should also be noted that the question of the optimality of hiring superstars is not fully answered and requires further research. For instance, I focused on the performance in the regular season, while the most important part of the season is the playoff for the Stanley Cup, so optimal decisions for these games should also be studied. Second, I have not taken into account the personalities of the coaches, but it is possible that some of them work better with superstars than others. Finally, I split the players into two groups, defenders and attackers, but one may want to focus on the smaller groups like left and right wingers, centers, defenders, and goalies. These further steps may complement my analysis and help researchers better understand which decisions regarding hiring superstars in the NHL are optimal.

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A Appendix: Theoretical Model

A.1 Algorithm for Solving the Maximization Problem

As I stated in the main part of my thesis, the function defined in (9) is not continuous, so I decided to find numerical solutions for given parameters. For this purpose, I created the script that implements the algorithm for finding the solution that I introduce in this subsection.

Now I describe how I find the optimal distribution for given α , β , $\tilde{w}$, w_{min} , w_{max} , and n .

1. The budget constraint specified in (1) is binding.

If it were not binding, the team could increase the salaries of the least paid players by a very small amount, which would decrease inequality and increase the first part of the productivity function. This would allow the teams to obtain a higher value for the productivity function. Therefore, the initial distribution could not be the one that maximizes the function. Hence, the budget constraint is binding.

This is a common result for all inequality functions, but I will continue using the HHI, but take into account the fact that the budget constraint is binding, so it is possible to use $g(w) = \sum_{i=1}^n w_i^2$.

2. Let's omit the constraints specified in (2) regarding the minimum and maximum wages for now.

Let's find out how the money should optimally be distributed among the players if there are n_{min} players getting the minimum wage (w_{min}), n_{max} players getting the maximum wage (w_{max}), k superstar players who have a superstar effect on others and are expected to get the wage in range $(\tilde{w}, w_{max})$, and $n - k - n_{min} - n_{max}$ non-superstar players who are expected to get the wage in range $(w_{min}, \tilde{w}]$. In this setting, the optimal wages for $n - n_{min} - n_{max}$ players should be determined, as the wages for $n_{min} + n_{max}$ players are already determined. Without loss of generality, let's assume that they are the players with numbers from 1 to $m = n - n_{min} - n_{max}$, and the first k of them are the superstars. Then, omitting the minimum and maximum wage constraints for these players, we have the following maximization problem:

$$F(w) = \sum_i w_i(1 + \beta \cdot \mathbb{I}\{w_i > \tilde{w}\}) - \alpha \sum_i w_i^2 + \lambda(\sum_i w_i - 1) \rightarrow \max_{w_1, \dots, w_m} \quad (12)$$

Then the first-order conditions are:

$$\forall i = 1, \dots, k : 1 + \beta - 2\alpha w_i + \lambda = 0$$

$$\forall i = k+1, \dots, m : 1 - 2\alpha w_i + \lambda = 0$$

The solution to this system is the following:

$$\forall i = 1, \dots, k : w_i = w_{sup} = \frac{1 - n_{min}w_{min} - n_{max}w_{max}}{n - n_{min} - n_{max}} + \frac{\beta(n - k - n_{min} - n_{max})}{2\alpha(n - n_{min} - n_{max})}$$

$$\forall i = k+1, \dots, m : w_i = w_{non-sup} = \frac{1 - n_{min}w_{min} - n_{max}w_{max}}{n - n_{min} - n_{max}} - \frac{\beta k}{2\alpha(n - n_{min} - n_{max})}$$

The second-order conditions are satisfied as the matrix of the second derivatives is just a diagonal matrix with -2α on the main diagonal. Hence, this distribution is the local extremum for given α , β , $\tilde{w}$, n_{min} , n_{max} , and k . Note that I omit the constraints regarding the minimum and maximum wages and do not check that superstars actually get a wage higher than the superstar threshold. I will check all these conditions in the next few steps.

3. In the next step, I consider all possible combinations of n_{min} , n_{max} , and k such that $n_{min} + n_{max} + k \in [0, n]$. For each of these combinations, I find the optimal distribution using the results of the previous step.
4. Then, for each combination, I calculate the value of the utility function that is achieved using the optimal distribution.
5. In the next step, I remove the distributions that do not satisfy at least one of the following constraints:

- Budget constraint:

$$n_{min}w_{min} + n_{max}w_{max} + kw_{sup} + (n - k - n_{min} - n_{max})w_{non-sup} = 1$$

- Individual constraints for superstars (if $k > 0$):

$$w_{sup} \in (\tilde{w}, w_{max}]$$

- Individual constraints for non-superstars (if $n - k - n_{min} - n_{max} > 0$):

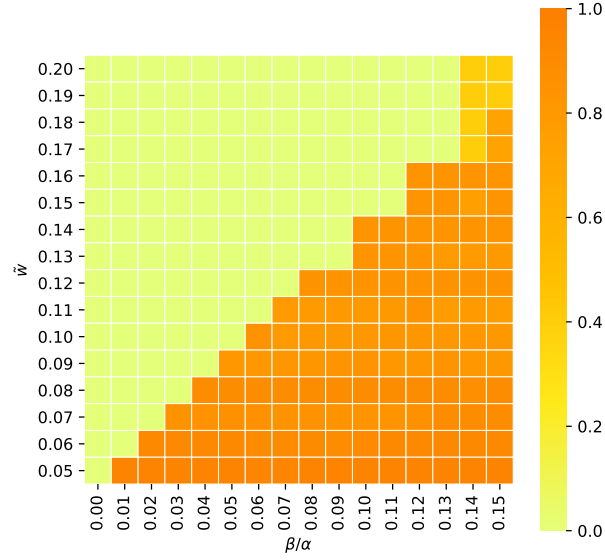
$$w_{non-sup} \in [w_{min}, \tilde{w}]$$

6. In the final step, I compare the distributions that satisfy all the conditions and choose the one that allows the team to achieve the maximum productivity. This distribution is the solution to the team's maximization problem. Note that there is always at least one distribution as $\forall i : w_i = \frac{1}{n}$ is one of the considered options and it satisfies all the constraints (under realistic assumption such that $n \cdot w_{min} < 1$).

After this step, the optimal distribution is calculated for given α , β , and $\tilde{w}$, so this algorithm can be run for different sets of parameters to make some predictions, as I do in the main part of my thesis.

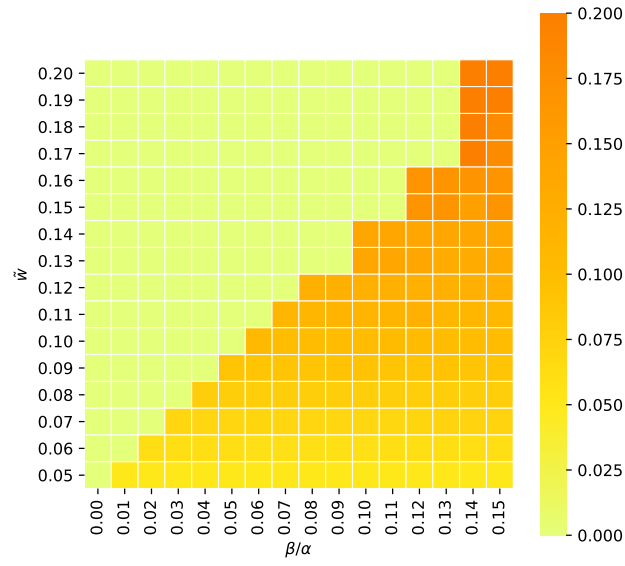
A.2 Additional Figures

Figure 4: Superstars' payroll in the optimal solution for given parameters



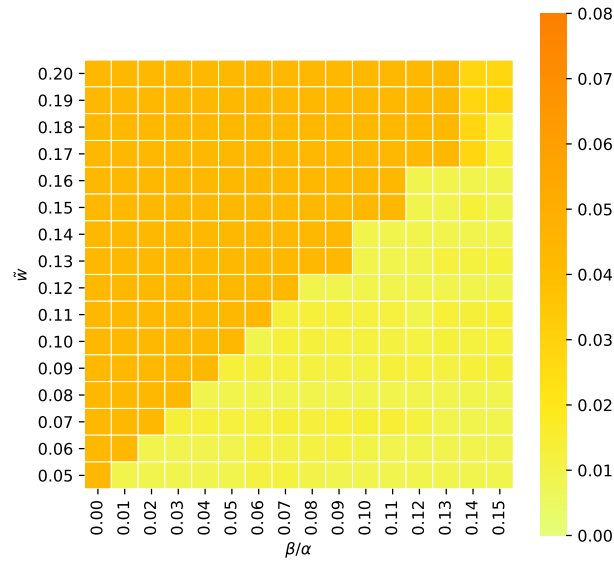
Notes: In the X-axis, I plot the superstar threshold $\tilde{w}$. In the Y-axis, I plot the ratio of the superstar effect to the inequality effect. The color of each square corresponds to the share of the total payroll that is spent on superstars with respect to the theoretical solution.

Figure 5: Average superstar salaries in the optimal solution for given parameters



Notes: In the X-axis, I plot the superstar threshold $\tilde{w}$. In the Y-axis, I plot the ratio of the superstar effect to the inequality effect. The color of each square corresponds to the average salaries that the superstars should get with respect to the theoretical solution.

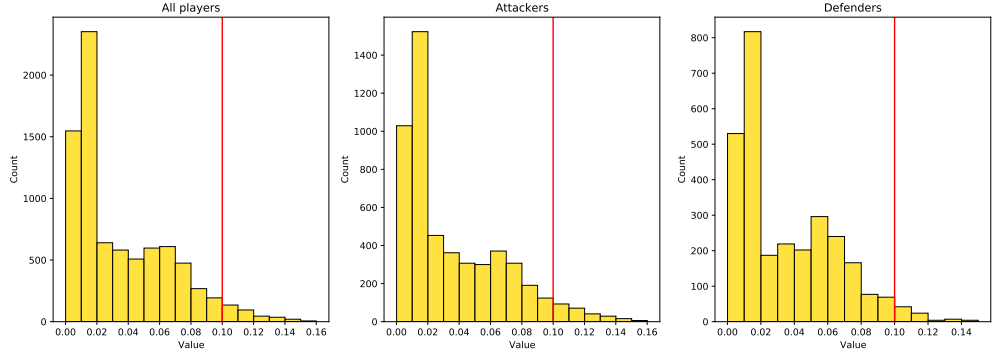
Figure 6: Average non-superstar salaries in the optimal solution for given parameters



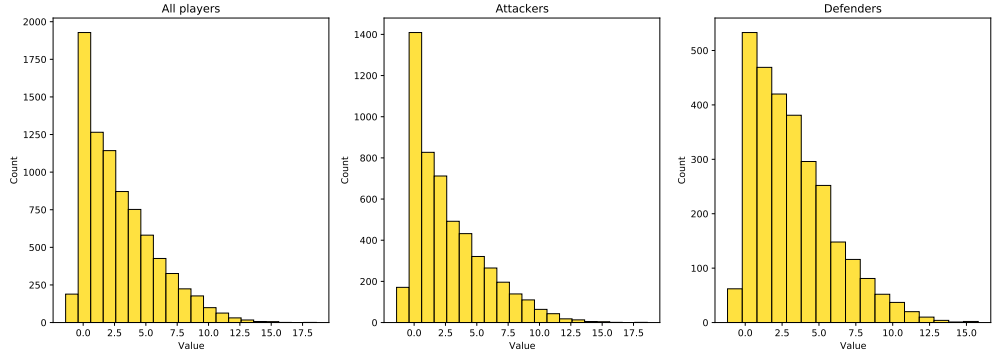
Notes: In the X-axis, I plot the superstar threshold $\tilde{w}$. In the Y-axis, I plot the ratio of the superstar effect to the inequality effect. The color of each square corresponds to the average salaries that non-superstar players should get with respect to the theoretical solution.

B Appendix: Data Description

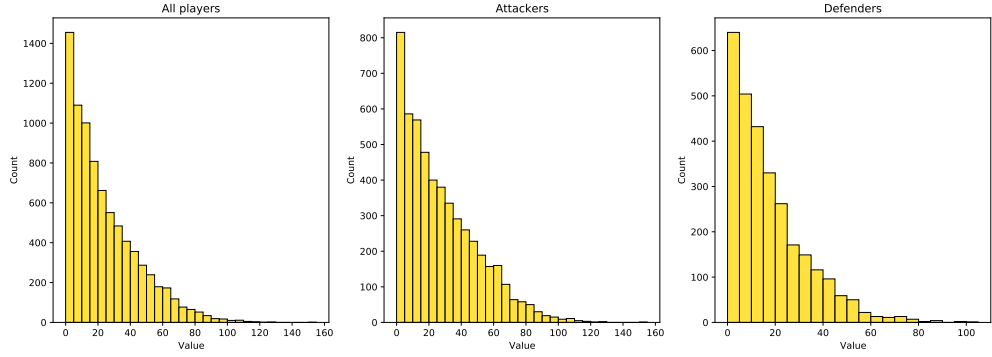
Figure 7: Histograms with distributions of player-level data



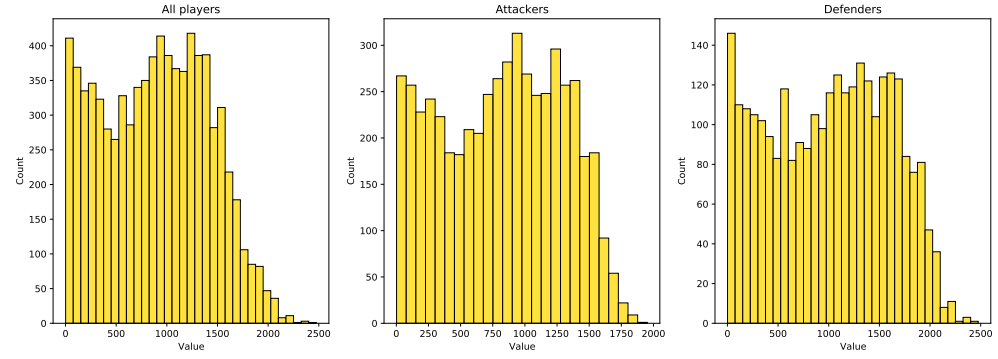
(a) Relative salary



(b) Points Share



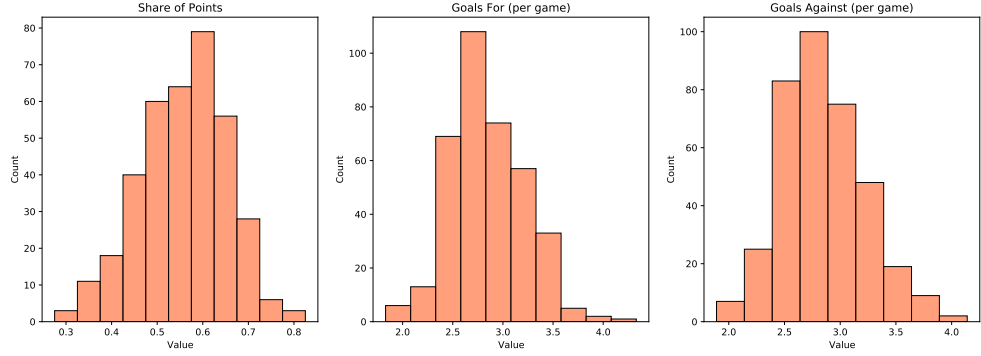
(c) Points (Goals and Assists)



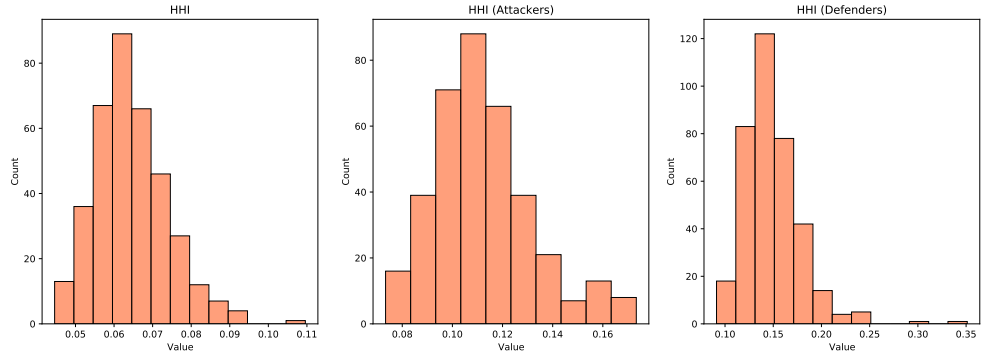
(d) Time on Ice

Notes: Data from [Sportrac.com](https://www.sportradar.com) for salaries and [Hockey-Reference.com](https://www.hockey-reference.com) for performances. Red lines in the first row stand for the threshold level of $\tilde{w} = 0.1$.

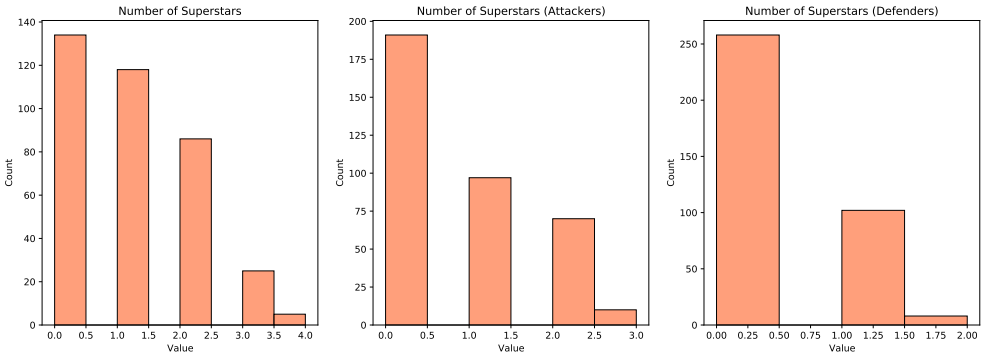
Figure 8: Histograms with distributions of player-level data



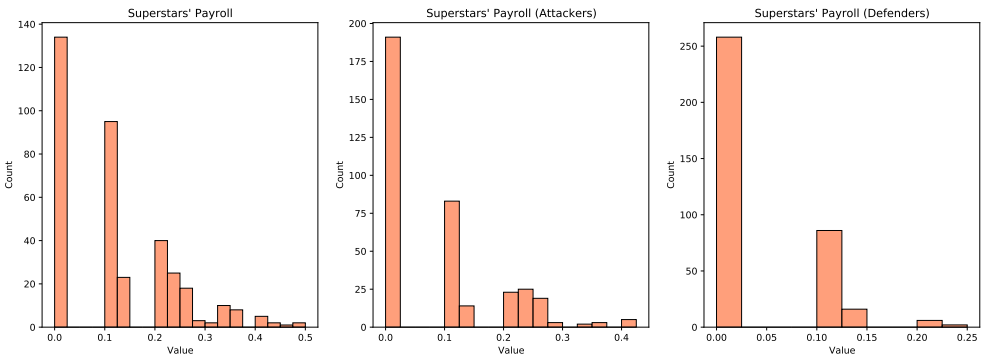
(a) Teams' Performance Measures



(b) HHI



(c) Number of Superstars



(d) Superstars' Payroll

Notes: Data from [Spotrac.com](https://www.spotrac.com) for salaries and [Hockey-Reference.com](https://www.hockey-reference.com) for performances. The superstars are the players getting more than 10% of the league's cap.

C Appendix: Additional Tables

Table 9: Analysis of the superstar and inequality effects using the player-level data (alternative functional forms)

VARIABLES	(1) ln(Points Share)	(2) ln(Points Share)	(3) ln(Points Share)	(4) ln(Points Share)	(5) ln(Points Share)	(6) ln(Points Share)
Relative salary	11.57*** (0.307)	11.57*** (0.307)	11.57*** (0.305)	11.56*** (0.308)	11.56*** (0.308)	11.57*** (0.308)
ln(HHI)	-0.262*** (0.0709)	-0.255*** (0.0705)	-0.256*** (0.0719)	-0.270*** (0.0701)	-0.264*** (0.0697)	
Superstars' payroll	0.283*** (0.0790)		0.0506 (0.579)	0.521*** (0.178)		0.277*** (0.0794)
Number of superstars		0.0327*** (0.00905)	0.0270 (0.0671)		0.0616*** (0.0202)	
(Superstars' payroll) ²				-0.720 (0.487)		
(Number of superstars) ²					-0.0105 (0.00664)	
HHI						0.103 (8.541)
HHI ²						-30.47 (63.27)
Constant	0.0947 (0.199)	0.115 (0.198)	0.111 (0.201)	0.0665 (0.197)	0.0823 (0.196)	0.940*** (0.282)
Team FE	Yes	Yes	Yes	Yes	Yes	Yes
Season FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	8,107	8,107	8,107	8,107	8,107	8,107
Number of teams	32	32	32	32	32	32
R-squared	0.329	0.329	0.329	0.329	0.329	0.329

Notes: Estimated coefficients are obtained using the TWFE model and data on all available players except goalies for seasons from 2011/12 to 2022/23. Data is collected from [Spotrac.com](https://www.spotrac.com) for salaries and [Hockey-Reference.com](https://www.hockey-reference.com) for performance measures. Salaries and payrolls are divided by the league's cap. Superstars are assumed to be the players who get salaries higher than 10% of the league's cap. The payrolls of superstars and non-superstars do not include the salary of considered individual. The performance measure, Points Share, is transformed using the function $f(x) = \ln(1.5 + x)$ to take into account the observations with negative values. Standard errors clustered at the team level are reported in parentheses. The p-values are denoted as follows: *** p<0.01, ** p<0.05, * p<0.1.

Table 10: Analysis of the superstar and inequality effects using the player-level data (alternative inequality measures)

VARIABLES	(1) Points Share	(2) Points Share	(3) Points Share	(4) Points Share	(5) Points Share	(6) Points Share
Relative salary	52.12*** (1.550)	51.24*** (1.576)	51.96*** (1.537)	51.41*** (1.571)	51.89*** (1.540)	51.26*** (1.558)
HHI	-18.51*** (4.323)	-15.74** (6.756)				
Gini			-3.497*** (1.069)	-2.931** (1.202)		
CV					-1.337*** (0.419)	-1.066** (0.504)
Superstars' payroll	0.897*** (0.322)	1.083*** (0.377)	0.982*** (0.308)	1.523*** (0.329)	1.007*** (0.305)	1.441*** (0.332)
D(Relative salary > 0.1)		0.304 (0.302)		0.314 (0.301)		0.320 (0.301)
Non-superstars' payroll		0.335 (0.508)		0.888** (0.391)		0.770* (0.413)
Constant	2.063*** (0.274)	1.614* (0.820)	2.420*** (0.473)	1.400* (0.762)	2.002*** (0.352)	1.114 (0.689)
Team FE	Yes	Yes	Yes	Yes	Yes	Yes
Season FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	8,107	8,107	8,107	8,107	8,107	8,107
Number of teams	32	32	32	32	32	32
R-squared	0.364	0.364	0.363	0.364	0.363	0.364

Notes: Estimated coefficients are obtained using the TWFE model and data on all available players except goalies for seasons from 2011/12 to 2022/23. Data is collected from [Spotrac.com](https://www.spotrac.com) for salaries and [Hockey-Reference.com](https://www.hockey-reference.com) for performance measures. Salaries and payrolls are divided by the league's cap. Superstars are assumed to be the players who get salaries higher than 10% of the league's cap. The payrolls of superstars and non-superstars do not include the salary of considered individual. Standard errors clustered at the team level are reported in parentheses. The p-values are denoted as follows: *** p<0.01, ** p<0.05, * p<0.1.

Table 11: Analysis of the superstar and inequality effects using the team-level data (alternative inequality measures)

VARIABLES	(1) Points (%)	(2) Points (%)	(3) Points (%)	(4) Points (%)	(5) Points (%)	(6) Points (%)
Relative payroll	0.259*** (0.0760)	0.269*** (0.0769)	0.449*** (0.0577)	0.449*** (0.0579)	0.431*** (0.0656)	0.433*** (0.0660)
HHI	-3.550*** (0.907)	-3.360*** (0.900)				
Gini			-0.242 (0.183)	-0.226 (0.187)		
CV					-0.105 (0.0790)	-0.0968 (0.0804)
Superstars' payroll	0.210*** (0.0686)		0.120* (0.0622)		0.131** (0.0629)	
Number of superstars		0.0232*** (0.00754)		0.0137* (0.00695)		0.0147** (0.00707)
Constant	0.522*** (0.118)	0.501*** (0.119)	0.234* (0.119)	0.228* (0.121)	0.232* (0.115)	0.223* (0.117)
Team FE	Yes	Yes	Yes	Yes	Yes	Yes
Season FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	368	368	368	368	368	368
Number of teams	32	32	32	32	32	32
R-squared	0.471	0.469	0.441	0.441	0.442	0.442

Notes: Estimated coefficients are obtained using the TWFE model and data on all teams for seasons from 2011/12 to 2022/23. Data is collected from [Spotrac.com](https://www.spotrac.com) for salaries and [Hockey-Reference.com](https://www.hockey-reference.com) for performance measures. Payrolls are divided by the league's cap. Superstars are assumed to be the players who get salaries higher than 10% of the league's cap. Standard errors clustered at the team level are reported in parentheses. The p-values are denoted as follows: *** p<0.01, ** p<0.05, * p<0.1.

Table 12: Analysis of the superstar and inequality effects using the player-level data (alternative superstar definitions)

VARIABLES	(1) Points Share	(2) Points Share	(3) Points Share	(4) Points Share	(5) Points Share	(6) Points Share
Relative salary	52.12*** (1.550)	52.11*** (1.549)	52.02*** (1.511)	51.95*** (1.505)	51.99*** (1.514)	51.98*** (1.514)
HHI	-18.51*** (4.323)	-18.07*** (4.305)	-16.28*** (4.087)	-15.68*** (4.057)	-17.65*** (4.191)	-17.57*** (4.191)
Superstars' payroll ($\tilde{w} = 0.1$)	0.897*** (0.322)					
Number of superstars ($\tilde{w} = 0.1$)		0.102** (0.0381)				
Superstars' payroll ($\tilde{w} = 0.08$)			0.345 (0.294)			
Number of superstars ($\tilde{w} = 0.08$)				0.0208 (0.0290)		
Superstars' payroll ($\tilde{w} = 0.12$)					0.795 (0.511)	
Number of superstars ($\tilde{w} = 0.12$)						0.103 (0.0702)
Constant	2.063*** (0.274)	2.038*** (0.273)	1.945*** (0.275)	1.943*** (0.275)	2.086*** (0.269)	2.081*** (0.269)
Team FE	Yes	Yes	Yes	Yes	Yes	Yes
Season FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	8,107	8,107	8,107	8,107	8,107	8,107
Number of teams	32	32	32	32	32	32
R-squared	0.364	0.363	0.363	0.363	0.363	0.363

Notes: Estimated coefficients are obtained using the TWFE model and data on all available players except goalies for seasons from 2011/12 to 2022/23. Data is collected from [Sportrac.com](https://www.sportrac.com) for salaries and [Hockey-Reference.com](https://www.hockey-reference.com) for performance measures. Salaries and payrolls are divided by the league's cap. Superstars are assumed to be the players who get salaries higher than 10% of the league's cap in columns (1) and (2), 8% in columns (3) and (4), and 12% in columns (5) and (6). The payrolls of superstars and non-superstars do not include the salary of considered individual. Standard errors clustered at the team level are reported in parentheses. The p-values are denoted as follows: *** p<0.01, ** p<0.05, * p<0.1.

Table 13: Analysis of the superstar and inequality effects using the team-level data (alternative superstar measures)

VARIABLES	(1) Points (%)	(2) Points (%)	(3) Points (%)	(4) Points (%)	(5) Points (%)	(6) Points (%)
Relative payroll	0.259*** (0.0760)	0.269*** (0.0769)	0.267*** (0.0745)	0.321*** (0.0796)	0.374*** (0.0656)	0.379*** (0.0664)
HHI	-3.550*** (0.907)	-3.360*** (0.900)	-3.081*** (0.882)	-2.530*** (0.903)	-2.297*** (0.779)	-2.224*** (0.785)
Superstars' payroll ($\tilde{w} = 0.1$)	0.210*** (0.0686)					
Number of superstars ($\tilde{w} = 0.1$)		0.0232*** (0.00754)				
Superstars' payroll ($\tilde{w} = 0.08$)			0.127** (0.0490)			
Number of superstars ($\tilde{w} = 0.08$)				0.00795* (0.00456)		
Superstars' payroll ($\tilde{w} = 0.12$)					0.0775 (0.104)	
Number of superstars ($\tilde{w} = 0.12$)						0.00848 (0.0141)
Constant	0.522*** (0.118)	0.501*** (0.119)	0.478*** (0.116)	0.404*** (0.122)	0.356*** (0.103)	0.348*** (0.104)
Team FE	Yes	Yes	Yes	Yes	Yes	Yes
Season FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	368	368	368	368	368	368
Number of teams	32	32	32	32	32	32
R-squared	0.471	0.469	0.457	0.451	0.447	0.447

Notes: Estimated coefficients are obtained using the TWFE model and data on all teams for seasons from 2011/12 to 2022/23. Data is collected from [Spotrac.com](https://www.sportradar.com) for salaries and [Hockey-Reference.com](https://www.hockey-reference.com) for performance measures. Payrolls are divided by the league's cap. Superstars are assumed to be the players who get salaries higher than 10% of the league's cap in columns (1) and (2), 8% in columns (3) and (4), and 12% in columns (5) and (6). Standard errors clustered at the team level are reported in parentheses. The p-values are denoted as follows: *** p<0.01, ** p<0.05, * p<0.1.

Table 14: Analysis of the superstar and inequality effects using the team-level data (alternative performance measures)

VARIABLES	(1) Points (%)	(2) Points (%)	(3) Goals For	(4) Goals For	(5) Goals Against	(6) Goals Against
HHI (A)	-0.811* (0.401)	-0.756* (0.406)	-1.102 (1.424)	-0.885 (1.424)	1.222 (1.279)	1.210 (1.266)
HHI (D)	-0.615** (0.238)	-0.636** (0.241)	-0.834 (0.872)	-0.904 (0.883)	1.835** (0.893)	1.914** (0.882)
Relative payroll (A)	0.183** (0.0702)	0.188** (0.0712)	0.372 (0.229)	0.402* (0.232)	-1.093*** (0.313)	-1.085*** (0.314)
Relative payroll (D)	0.359*** (0.0954)	0.353*** (0.0933)	1.291*** (0.306)	1.270*** (0.309)	-0.881** (0.348)	-0.851** (0.342)
Superstars' payroll (A)	0.207*** (0.0720)		0.415 (0.287)		-0.315 (0.237)	
Superstars' payroll (D)	-0.0222 (0.101)		-0.134 (0.356)		0.0644 (0.404)	
Number of superstars (A)		0.0244*** (0.00855)		0.0452 (0.0347)		-0.0399 (0.0277)
Number of superstars (D)		-0.000582 (0.00998)		-0.00788 (0.0373)		-0.00139 (0.0413)
Constant	0.480*** (0.0939)	0.476*** (0.0937)	2.346*** (0.342)	2.324*** (0.331)	3.414*** (0.366)	3.393*** (0.358)
Team FE	Yes	Yes	Yes	Yes	Yes	Yes
Season FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	368	368	368	368	368	368
Number of teams	32	32	32	32	32	32
R-squared	0.498	0.499	0.558	0.558	0.538	0.539

Notes: Estimated coefficients are obtained using the TWFE model and data on all teams for seasons from 2011/12 to 2022/23. Data is collected from [Spotrac.com](https://www.spotrac.com) for salaries and [Hockey-Reference.com](https://www.hockey-reference.com) for performance measures. Payrolls are divided by the league's cap. Superstars are assumed to be the players who get salaries higher than 10% of the league's cap. "A" in the parentheses stands for Attackers, while "D" is for Defenders. Standard errors clustered at the team level are reported in parentheses. The p-values are denoted as follows: *** p<0.01, ** p<0.05, * p<0.1.

Table 15: Analysis of the superstar and inequality effects using the team-level data (splitting the dataset by relative payrolls)

VARIABLES	(1) Points (%) All teams	(2) Points (%) All teams	(3) Points (%) Payroll > 0.9	(4) Points (%) Payroll > 0.9	(5) Points (%) Payroll > 0.95	(6) Points (%) Payroll > 0.95
Relative payroll	0.259*** (0.0760)	0.269*** (0.0769)	0.411* (0.240)	0.421* (0.237)	0.430 (0.563)	0.402 (0.566)
HHI	-3.550*** (0.907)	-3.360*** (0.900)	-4.385*** (1.203)	-4.166*** (1.130)	-3.908** (1.514)	-3.850** (1.421)
Superstars' payroll	0.210*** (0.0686)		0.216*** (0.0747)		0.202** (0.0823)	
Number of superstars		0.0232*** (0.00754)		0.0246*** (0.00817)		0.0246*** (0.00884)
Constant	0.522*** (0.118)	0.501*** (0.119)	0.428 (0.280)	0.405 (0.274)	0.379 (0.614)	0.401 (0.614)
Team FE	Yes	Yes	Yes	Yes	Yes	Yes
Season FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	368	368	249	249	204	204
Number of teams	32	32	30	30	30	30
R-squared	0.471	0.469	0.352	0.352	0.343	0.348

Notes: Estimated coefficients are obtained using the TWFE model and data on all teams for seasons from 2011/12 to 2022/23. Data is collected from [Sportrac.com](https://www.sportrac.com) for salaries and [Hockey-Reference.com](https://www.hockey-reference.com) for performance measures. Payrolls are divided by the league's cap. Superstars are assumed to be the players who get salaries higher than 10% of the league's cap. Standard errors clustered at the team level are reported in parentheses. The p-values are denoted as follows: *** p<0.01, ** p<0.05, * p<0.1.